

1(a)(i)	direction of the force acting on a (test) mass placed at the point	B1
1(a)(ii)	change in height negligible compared with radius (of Earth)	B1
	(so) field lines are (effectively) parallel	B1
1(b)(i)	$Y = GM / R^2$	M1
	G is the gravitational constant	A1
1(b)(ii)	gravitational force is (always) attractive or gravitational force (always) acts towards the centre of the sphere	B1
	force is in opposite direction to displacement or at a point to the right of the centre, force acts to the left or at a point to the left of the centre, force acts to the right	B1
1(b)(iii)	sketch: smooth curve with decreasing positive gradient, starting at $(R, -Y)$ and reaching $3R$ with g still negative or smooth curve with increasing positive gradient, ending at $(-R, Y)$ and reaching $-3R$ with g still positive	B1
	both of the above curves, in correct quadrants	B1
	curve passing through $(\pm 2R, \pm 0.25Y)$ and $(\pm 3R, \pm 0.11Y)$	B1